Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Lorell, Beverly H M.D.		
Project Title:	Cardiac Unloading: Biologic Mechanisms and Countermeasures for Cardiac Atrophy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:	Co-Investigator: Michael D. Schneider, M.D. Baylor College of Medicine Houston, Texas 77030-3498 Collaborators: Thomas K. Borg, Ph.D. and Robert J. Price, Ph.D. Dept of Developmental Biology and Anatomy University of South Carolina Columbia, South Carolina 29208		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Lorell, Beverly		
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-CA.002.04		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	GOALS. The project objective is to determine the cellular and molecular mechanisms of atrophy induced by microgravity (already demonstrated in space-flown rats and long-duration spaceflight in humans), determine the functional consequences on cardiac contractile reserve, and identify specific countermeasures relevant to human intervention. Specific Aims are: 1) Are ventricular myocyte contractile function and intracellular Ca2+ regulation modified in cardiac unloading and atrophy? 2) Does cardiac unloading stimulate programmed cell death (apoptosis) in addition to remodeling of myocyte morphology? 3) Do hormonal and genetic countermeasures blunt atrophy in vivo and/or favorably modify functional impairments? APPROACHES. The Cardiovascular Alterations Team was required to develop surrogate earth-based models of microgravitational unloading, in anticipation of potential later experimentation in space-flown experiments. This project utilizes a well-established rodent model of cardiac unloading (heterotopic transplantation of the heart to the abdomen) which affords analysis of progressive degrees and magnitude of unloading in a compressed timespan, compared with human lifespan. Functional consequences of unloading, and effects of countermeasures, are studied by measurements of isolated adult myocyte contraction and simultaneous regulation of intracellular ions using high speed video and fluorescence microscopy, and microscopy of cell morphology. The activity of key endogenous regulators of contractility and growth are monitored by measurements of cardiac gene message and protein levels and immunohistochemical localization. High-throughput profiling of gene expression using microarray and subtractive hybridization approaches will complement this approach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The insights developed from this project are synergistic with a major public health problem in the field of heart disease: remodeling and dysfunction of the heart associated with changes in load. Our findings to date have this impact on understanding and management of clinical hypertrophy and failure in patients: 1) Normal adult cardiac gene programming appears to operated within very narrow upper and lower boundaries of load; 2). Prolonged excess unloading promotes changes in gene reprogramming that decrease performance reserve of the heart. This is directly relevant to patient treatment strategies in heart failure, including prolonged unloading due to LV assist devices.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)